



Nitrogen Oxidation on A Plasma-Exposed Surface

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Abstract

The elementary processes during the fixation of nitrogen by plasma catalysis are studied in a low-pressure plasma experiment with N_2 and O_2 as source gases. The formation of surface groups on an iron oxide foil is monitored with infrared reflection absorption spectroscopy. Surface nitrates (NO_3^-) are formed when the substrate is exposed to a 1:1 $N_2:O_2$ plasma, as well as $N_2O(g)$, $NO(g)$, $NO_2(g)$, and $O_3(g)$ in the gas phase. It is postulated that $NO_{1,2}(g)$ species created by the plasma, adsorb at the surface and create these nitrates. This constitutes an intermediate step for nitrogen oxidation by plasma catalysis.

Keywords N_2/O_2 plasma · Surface composition · Plasma catalysis · RF plasma

Introduction

Plasmas are a viable tool for species conversion using non-equilibrium synthesis routes, which may exhibit high energy and mass efficiency. This motivates many groups to explore various chemical systems, ranging from CO_2 splitting, ammonia synthesis, methane pyrolysis, to NO_x formation [1–3]. It is postulated that the combination of plasma with catalysis may enhance the selectivity of these processes. Based on microkinetic modelling, the combination of plasma and catalysis may also lead to a large enhancement of reaction rates and a shift to lower required surface temperatures compared to thermal catalysis [4]. However, these theoretical predictions of large enhancements could not be confirmed. Instead, the contribution of radicals seems to dominate a plasma catalysis system so that any activation barriers for molecules at the surface are no longer defined as the rate-limiting step [5]. The successful application of plasma catalysis requires a much better understanding of how the surface species at a catalyst change during plasma operation.

The analysis of surface processes is usually performed at ultra-high vacuum conditions using electron-based spectroscopies [6, 7]. At these conditions, the molecules of interest are dosed to well-defined and characterized surfaces, where the contribution of any impurities is minimal. Such surface science experiments provide very valuable information on the kinetics and thermodynamics of such reactions. Still, the initial species are almost exclusively stable molecules, whereas radicals are provided instead in

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a plasma experiment. Consequently, surface processes relevant to plasma catalysis must be probed in operando. This limits the experimental methods to non-invasive optical methods that can be operated simultaneously with a plasma. One possibility is to probe the surface with infrared reflection absorption spectroscopy (IRRAS). This diagnostic measures the vibrational transitions of molecules adsorbed on the surface and in the gas phase. Identifying IR lines in gas phase spectra is straightforward, but identifying surface groups is more ambiguous.

In this paper, we explore the plasma catalysis synthesis routes of NO_x species. The oxidation of nitrogen is well studied, and ample literature exists for species identification, e.g. see [8]. The adsorption of NO on an iron surface forms mono-nitrosyl Fe-(NO), di-nitrosyl Fe-(NO)₂, and NO⁺ [9]. Nitrates (NO₃⁻) are formed if oxygen is added during or after NO exposure [9–11]. This is similar to the adsorption of NO₂(g) on aluminium oxide [12]. Moreover, the reduction of NO surface species towards N₂O(g) is well understood [13]. All these measurements were typically performed at low pressures for varying surface temperatures and using stable molecules as incident species. This is, however, in stark contrast to environments created by plasmas.

Plasmas create a highly reactive environment. Radicals in the gas phase likely dominate the surface chemistry through Eley-Rideal reactions [14]. This potentially lifts the energy barrier for nitrogen oxidation processes and desorption reactions, thereby potentially allowing NO_x formation at very low surface temperatures. In addition, ions impinge on the surface as well, which are accelerated by the plasma sheath. These impinging ions may change the surface groups due to collision processes or due to the release of electronic energy when they neutralize upon impact. Yet, these effects are likely overshadowed by the high radical flux. It is, therefore, interesting to understand how nitrogen oxidation occurs via plasma catalysis and how it differs from thermal oxidation.

The main aim of the experiment is to monitor surface changes upon direct plasma impact. To understand the changes on the surface, we need to detect both surface-bound and gas-phase species with high sensitivity. As a result, we choose IRRAS, where both species are probed by the IR beam. The high stability of our setup allows us to observe changes in the absorption of 0.01%, thereby allowing low surface coverages and gas phase densities to be detected. Due to the intermediate pressure plasma operation and the weak absorption signal, one must compete with the impurities in the system. They are unavoidable because they are either introduced by the gas supply or by the reaction of the reactive plasma with the chamber walls. Any dedicated pretreatment of the sample surface, as it is performed in a typical thermal catalysis experiment, is not mandatory. The plasma, as a reactive environment, will create its own reactive surface due to the supply of reactive oxygen or nitrogen species. This is very different to a thermal catalysis system. There, a catalyst can easily be poisoned, rendering it inactive towards stable molecules. When starting a plasma experiment, the very first infrared spectra might show influences of impurities on the surface chemistry. This influence will disappear in the subsequent spectra, since the influx of reactive species dominates the surface processes. Nonetheless, their influence can be minimised by removing them in previous steps. In summary, our experiment monitors with great sensitivity a plasma-exposed surface. The state of the surface is determined largely by the plasma and not so much by the pretreatment before plasma operation.

In this study, we characterise the composition of an iron foil that is in direct contact with a N₂:O₂ plasma. The composition is probed by IRRAS. A capacitive coupled plasma is ignited at a low pressure (8 mbar) using a radio frequency sinusoidal waveform. This creates a homogeneous cold plasma that is in direct contact with the iron foil. The

measured reflectance spectra are analyzed by comparing the features to the literature. Iron is used because it is known as a thermal catalyst. Hence, many literature sources on surface groups bound to iron surfaces can be found.

Methodology

Experimental Setup

A schematic of the setup is shown in Fig. 1a. The plasma is ignited between two rectangular electrodes in a planar reactor configuration, see Fig. 1b. The distance of the electrodes is 10 mm to allow the IR optical configuration, see Fig. 1b. This sets the plasma volume to $10 \times 10 \times 20$ mm. An iron foil (0.15 mm thickness, 99.5% purity) is placed on the grounded bottom electrode.

The plasma is ignited in a low-pressure compartment. This is well approximated as a cylinder with a diameter and height of 30 cm with sidearms of approximately 5 cm long. The gas flow is controlled with mass flow controllers. The reactor is filled with a desired gas mixture combining helium, nitrogen, and oxygen. Helium is used as a carrier gas during surface cleaning steps. Flow rates are set to 200 sccm for He and up to 20 sccm for N_2 and O_2 . The pressure is monitored by a capacitance gauge and manually controlled to 8 mbar by a rotating valve between the low-pressure compartment and the scroll pump. The temperature of the bottom electrode is monitored with a thermocouple. It increases up to 40° during continued plasma operation, thus the heating is presumed to be negligible in the surface kinetics.

A sinusoidal waveform is generated by an RF generator (RFG150-13, Coaxial Power Systems). The impedance is matched by a Manual Impedance Matcher. The voltage and current are measured by an oscilloscope and an in-house-made IV-box, which is also used in previous studies [15, 16].

The reflectance spectra are obtained with a Fourier Transform infrared (FTIR) spectrometer (Vertex 70v, Bruker) with a spectral resolution of 2 cm^{-1} . The light is focused

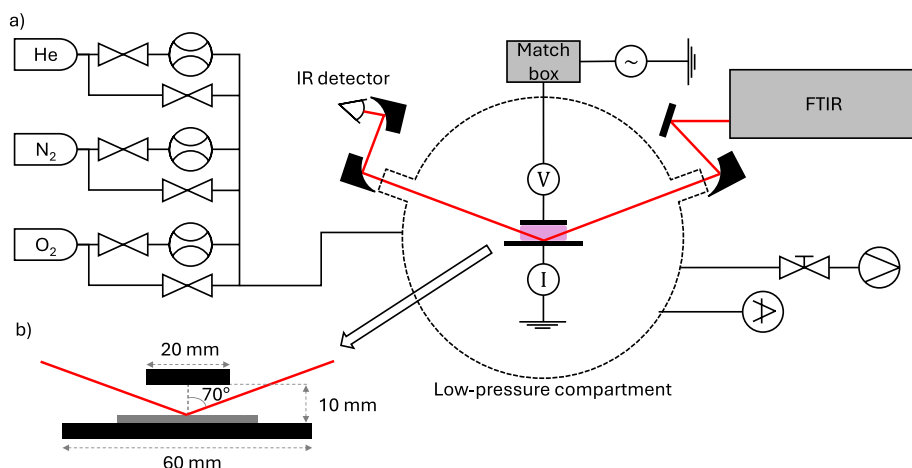


Fig. 1 A sketch of the setup of the gas lines, IR beam path (red line), and voltage and current measurements (color figure online)

on the sample using two protected gold-coated off-axis parabolic mirrors with a focal length of 250 mm (II-VI GmbH). Two KBr windows provide optical access to the reactor while isolating the compartment from the environment. The light is measured using an external mid-IR detector. The open-air path between the FTIR, reactor, and detector is purged with dry air to minimize IR light absorption by $\text{H}_2\text{O}(\text{g})$, which is naturally present in the air. The absorption by $\text{CO}_2(\text{g})$ remains.

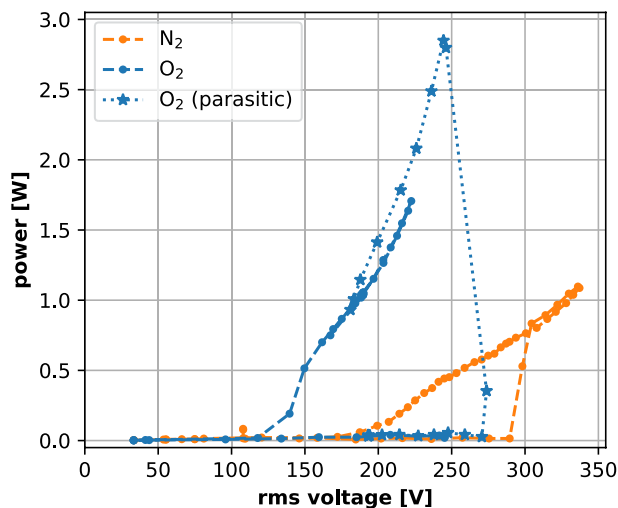
Plasma Operation

The power-to-voltage curves are measured during plasma operation as shown in Fig. 2 for a pure N_2 and O_2 plasma. This is useful to estimate the ignition mode in which the NO_x formation is studied, where the pure N_2 and O_2 plasma acts as extremes. The voltage is increased until the plasma ignites before a plasma power scan is performed. The power scales quadratically with the voltage, as is expected for a non-expanding plasma. This is best seen for the O_2 plasma, and for N_2 above 240 V. When igniting an N_2 plasma, the plasma contracts below 240 V and the slope of the curve changes. In addition, a parasitic discharge can be observed in an O_2 plasma if the plasma is ignited at very large voltages. However, a regular quadratic power scan can be reached if the voltage is reduced after ignition. In summary, plasma power is limited to values around 1 W, so a homogeneous plasma is obtained irrespective of the used plasma gas.

Data Acquisition

The IRRAS spectra are plotted as absorbance $A = -\ln I/I_{\text{ref}}$ vs. wavenumber. The reference spectrum I_{ref} is obtained before plasma ignition, and the spectrum I during the experiment. The absorbance contains features of both the surface and the gas phase. A change in the reflectance of the surface indicates a change in the surface composition [17]. The formation of surface groups increases with increasing absorbance; and vice-versa for removal. The IR light interacts with the top layer of the iron foil following the skin depth.

Fig. 2 The power versus the applied rms voltage for pure N_2 and O_2 (color figure online)



This is sub-nm for pure iron. However, the topmost layers of the samples in our experiment remain oxidised by the O₂ containing plasma. This yields a much larger probing depth in the range of the order of μm .

The gas phase absorption features are readily identified by comparing the observed absorption features to a synthetic gas phase spectrum. These spectra are simulated using the HITRAN database [18]. The corresponding densities are derived from fitting these synthetic spectra to the measured spectra. These spectra are assumed to be in rovibrational equilibrium since only a small part of the total absorption path length ($\approx 42.6\text{ cm}$) consists of the plasma itself ($\approx 2.1\text{ cm}$). In addition, the nature of our discharge does not heat up the gas, as mentioned before; thus, we simulated these spectra at room temperature. The density distribution is approximated to be homogeneous. This is validated by estimating the characteristic diffusion time from the plasma towards the edges of the reactor, which is well below the time resolution of the FTIR measurement [19, 20].

Finally, the IR beam path contains distinct absorption features of H₂O(g) and CO₂(g) in the gas phase, which originate from the beam path outside the low-pressure compartment. Their contribution in the spectra is minimised by scaling a reference spectrum with a concentration factor. The H₂O(g) reference spectrum is calculated using the HITRAN database. The contribution of CO₂ is removed using a measured spectrum. The observed CO₂(g) absorption is not easily reproduced. CO₂ is a very strong absorber, and details of the instrumental lineshape do not follow a simple Gaussian when using a coarse spectral resolution.

Sample Preparation

The analysis of surface processes requires a reliable initial reference state I_0 , since the pristine iron samples still contain undefined surface contaminations. In thermal catalysis experiments, the substrate is carefully pretreated before any experiments are performed. For instance, by pretreating the substrate in a O₂/He mixture at high temperatures [21]. Similar to that, we implemented a pretreatment protocol to identify and minimise the impurity IR absorption features. This simplifies the analysis of the IRRAS spectra.

The iron foils are cleaned by operating a 200:20 sccm He:O₂ plasma for 1 h. The addition of helium enhances the throughput whilst lowering the required voltage to operate the plasma, which reduces the heating of the system. Also, it contributes to the cleaning process by physical sputtering. Unlike hydrogen and oxygen, it does not chemically react with iron.

The He+O₂ plasma removes carbon impurities, while oxidising the top layer of the iron foil. This is evident from the IRRAS spectra in Fig. 3a, where the spectra that are obtained before igniting this plasma (0 min, in blue) and after 1 and 61 min of plasma operation (red) are presented. The identification of the lines is indicated using vertical grey dashed lines. The carbon impurity removal is already apparent after 1 min plasma exposure. This is evident from the CH₃ stretch at 2850;2920;2960 cm⁻¹ and C=O stretch at 1730 cm⁻¹ [22–24]. The fingerprint region 1000–1700 cm⁻¹ peaks are, however, generally more complicated to identify. Nonetheless, the peaks agree well with the expected bending modes of hydrocarbon and carbonyl groups [25–27].

After operating the plasma for 1 h, the previously mentioned removal features are stronger, i.e. more is removed. The carbonyl band at 1730 cm⁻¹ seems to blue shift to 1750 cm⁻¹. Also, the formation of Fe(III)oxide is observed around 720 cm⁻¹ [28–31]. The blue shifting of the C=O band either arises from the reforming of pre-existing

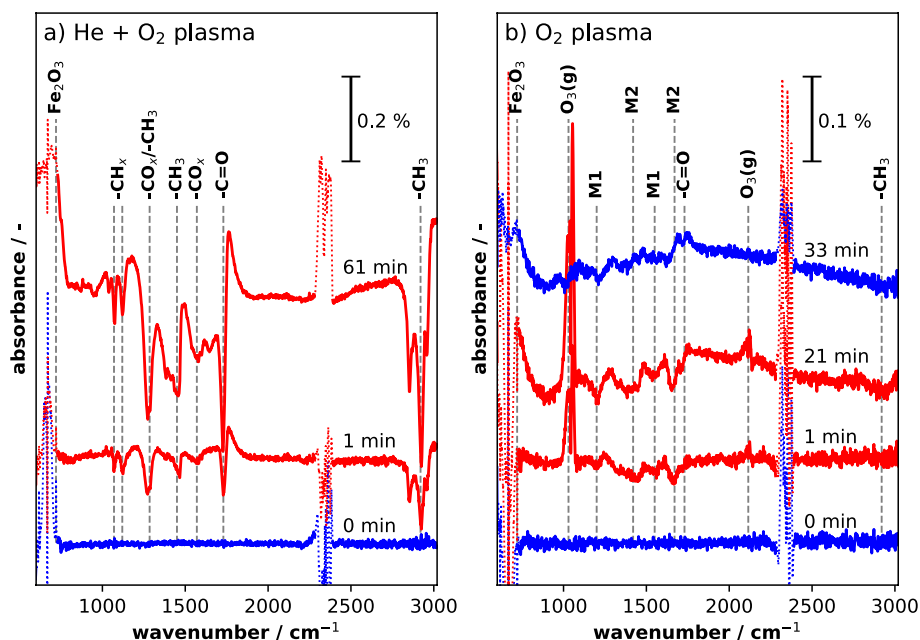


Fig. 3 The spectra obtained when igniting (red) and not igniting (blue) the He+O₂ (a) and pure O₂ plasma (b). Noteworthy absorption features are labelled with the corresponding species. The spectral region obscured by CO₂(g) is plotted with a dotted line (color figure online)

bonds or from the formation of an oxide film that introduces a phase shift. This means that despite the pretreatment, impurities persist during the various plasma operations. By a careful analysis of the spectra, however, their contribution can be taken into account. Moreover, they re-adsorb given enough time. Therefore, the plasma operations are done in quick succession after each other.

The substrate is further treated using a 20 sccm O₂ plasma for 20 min, see Fig. 3b. The changes in the spectra are less significant; therefore, the y-axis is enlarged by a factor of 2. Spectra before, during, and after the plasma treatment are shown. This plasma treatment leads to the oxidation of the iron foil, which is deduced from the Fe₂O₃ O₃ peak at 720 cm⁻¹. Also, O₃(g) is observed at 1030 and 2100 cm⁻¹. The previously observed carbonyl and hydrocarbon features are not observed, as the strong IR active lines at 1730 and around 2900 cm⁻¹ are absent in the spectra. Yet, minor changes in the fingerprint region of the spectrum are still observed when igniting the plasma. These are labeled as M1 and M2 and can easily be ascribed to carbonate, formate, etc. [23, 26]. Their origin is likely related to a minor residual contribution of carbon impurities. These features recover once the plasma is switched off.

Finally, the pretreatment removes carbon impurities from the spectra such that they are not observed in the absorbance spectra anymore. This does not necessarily mean that all carbon impurities are completely removed. For instance, the C=O stretch at 1730 cm⁻¹ blue shifts after continued He+O₂ plasma operation. Also, during the O₂ plasma, impurities are temporarily removed during plasma operation and then re-adsorb after the plasma is switched off. Therefore, we have to conclude that impurities are

inherently present on our initial substrate, but are removed during plasma treatment. Nonetheless, their contribution is minimised by these pretreatment steps and performing the experiments in quick succession. This greatly simplifies the analysis of the absorbance spectra, thereby enabling the identification of NO_x surface groups.

Results

In a first set of experiments, nitrogen oxidation on the surface is studied in an $\text{N}_2 + \text{O}_2$ plasma, which also creates gas phase $\text{NO}_x(\text{g})$. Therefore, the surface composition is expected to be similar to typical thermal oxidation processes, where stable $\text{NO}_x(\text{g})$ species are applied as well. In a second set of experiments, the surface composition is studied when cycling between pure N_2 and O_2 plasmas. Under those conditions, only pre-adsorbed oxygen/nitrogen can result in the formation of nitrogen oxides.

$\text{N}_2:\text{O}_2$ Plasma

Figure 4a shows the IRRAS spectra during exposure of the substrate to a 1:1 $\text{N}_2:\text{O}_2$ plasma, using a 10 sccm $\text{N}_2 + 10$ sccm O_2 gas flow. The *plasma-off* spectra are plotted in blue and the *plasma-on* spectra in red; similar to before. The first spectrum (0.0 min) is obtained

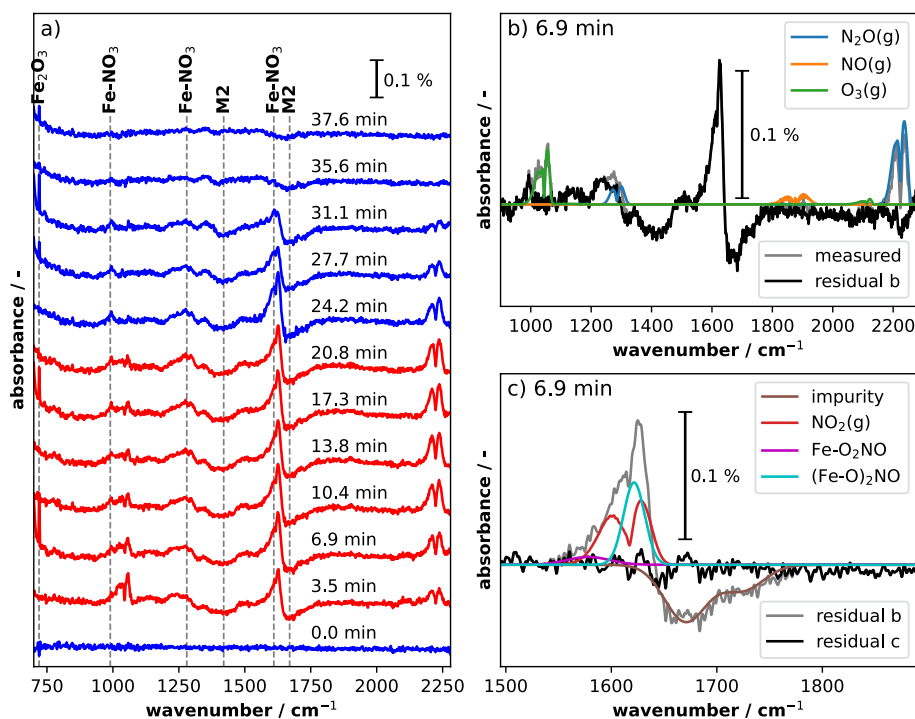


Fig. 4 IRRAS spectra in time (a) when the plasma is on (red) and off (blue) with a 10 sccm N_2 and 10 sccm O_2 gas flow, where blue indicates that the plasma is off and red is plasma on. The spectrum at 6.9 min is decomposed in (b) and (c)

before plasma ignition. The plasma was operated from 1.2 to 22.5 min, see red spectra. These spectra show the formation of both surface groups and gas species. Afterwards, the plasma is switched off, see the blue spectra from 24.2 min. These spectra show the gradual disappearance of these absorption features. The gas flow is stopped, and the reactor is evacuated from 32.3 min onward. In the end, the spectrum reverts to a horizontal line. This indicates the reversal of all plasma-induced processes.

The spectral positions of the absorption features reveal the changes in the chemical composition of the substrate. Peaks around 990 cm^{-1} , from 1200 to 1300 cm^{-1} , and around 1620 cm^{-1} are associated to nitrates [8, 32]. The latter is often attributed to either nitrites (NO_2^-) or nitrates (NO_3^-) species [9, 11, 33–35]. The close proximity between the vibrational bands makes it difficult to differentiate between them [11, 36–38]. Nonetheless, the majority associates it with nitrates. Different binding configurations lead to small shifts with respect to the central wavenumber. The deconvolution reveals two contributions: chelating bidentate nitrate $\text{Fe-O}_2\text{NO}$ at 1575 cm^{-1} and bridging bidentate nitrate ($\text{Fe-O}_2\text{NO}$) at 1620 cm^{-1} . These are well documented in literature [8–11, 32–38]. Therefore, we will use these bidentate nitrate bands to characterise the nitrate surface coverage.

Alternatively, adsorbed H_2O also has a vibrational band around 1600 cm^{-1} [39]. Yet, this should appear along with the OH stretch vibration around 3600 cm^{-1} that is absent in our measurements (not shown). Hence, this possibility can be excluded.

Furthermore, the absence of absorption peaks between 1700 and 2200 cm^{-1} rules out the presence of NO, N_2O_3 , and N_2O_4 on the surface [40–44]. This indicates that only nitrates are formed.

The spectra are separated into individual peaks to estimate the surface coverages and gas phase densities. This is performed in two steps. Figure 4b and c illustrates the analysis for a typical spectrum (6.9 min). The rotational structures of $\text{N}_2\text{O(g)}$, NO(g) , $\text{NO}_2\text{(g)}$, and $\text{O}_3\text{(g)}$ are used together with the bidentate nitrate peaks.

$\text{N}_2\text{O(g)}$, NO(g) , and $\text{O}_3\text{(g)}$ are spectrally isolated such that their contributions can be estimated independently. The N-N stretch of $\text{N}_2\text{O(g)}$ around 2220 cm^{-1} is fitted, leading to a prediction of the associated N-O stretch rovibrational band around 1280 cm^{-1} . NO(g) is observed around 1880 cm^{-1} . This molecule is less IR active as the other products and its absorption is in the same order as the noise. Nonetheless, its absorption band is fitted to the data. The $\text{O}_3\text{(g)}$ absorption around 1040 cm^{-1} is fitted. The overtone of $\text{O}_3\text{(g)}$ around 2110 cm^{-1} is calculated following the outcome of this fit, similar to the calculation of the N-O stretch from the signal of $\text{N}_2\text{O(g)}$. In the end, these contributions are subtracted from measured spectrum, which yields residual *b* spectrum.

In Fig. 4c, the absorption around 1620 cm^{-1} is decomposed. It consists of the bands of $\text{NO}_2\text{(g)}$, nitrate contributions $\text{Fe-O}_2\text{NO}$ and $(\text{Fe-O})_2\text{NO}$, and the removal of M2 impurities, which are mentioned in Sect. 2.4. First, the impurities are corrected for. This is approximated using two Gaussian peaks centred at 1670 and 1725 cm^{-1} with $\sigma = 20\text{ cm}^{-1}$, see impurity spectrum. Their contribution is fit to the data $> 1650\text{ cm}^{-1}$ and subtracted from residual *b*. Afterwards, the contributions of $\text{NO}_2\text{(g)}$ and the nitrates are fitted. The nitrate peaks are approximated with a Gaussian line shape with a broadening constant σ of 10 cm^{-1} . The presence of $\text{NO}_2\text{(g)}$ is experimentally verified by reducing the pressure from 8.0 to 0.4 mbar, thereby experimentally eliminating the IR signature of all gaseous species. The difference between the before and after spectra showed a clear contribution of $\text{NO}_2\text{(g)}$ (see supplementary information). Finally, these contributions are subtracted from spectrum residual *b* as well that gives spectrum residual *c*.

In Fig. 5, the evolution of the gas densities and nitrate peak intensities is plotted. The time domain is divided into 3 phases: plasma on (phase I), plasma off (phase II),

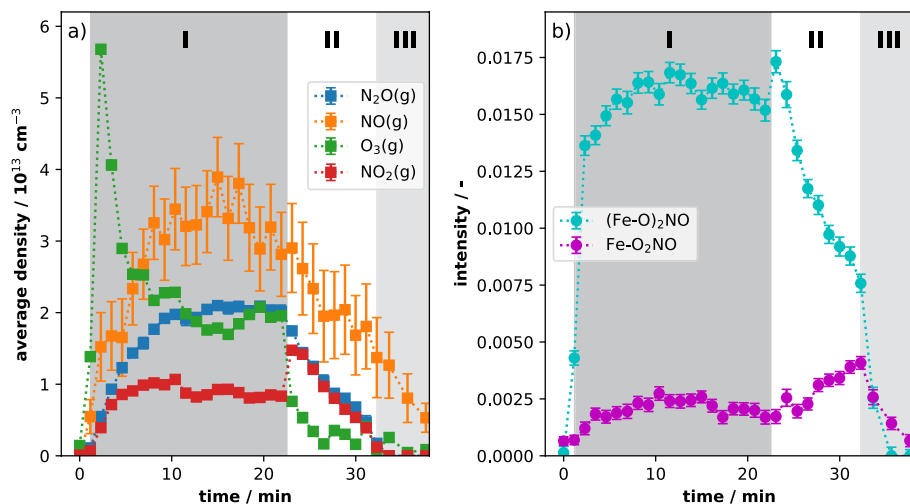


Fig. 5 The densities of $\text{O}_3(\text{g})$, $\text{N}_2\text{O}(\text{g})$, $\text{NO}_2(\text{g})$, and $\text{NO}(\text{g})$ (a) and the intensities (b) and spectral shift (c) of $\text{NO}_3(\text{s})$ peaks A, B, and C during plasma operation (phase I), in the plasma-off period (phase II), and when evacuating the reactor (phase III)

and evacuated (phase III). The presented densities are an average over the IR beam path inside the low-pressure compartment. The uncertainties are estimated from the fitting. On average, the uncertainties of $\text{N}_2\text{O}(\text{g})$, $\text{NO}(\text{g})$, $\text{O}_3(\text{g})$, and $\text{NO}_2(\text{g})$ densities are $7e11; 7e12; 7e11; 3e11 \text{ cm}^{-1}$, respectively. The intensities, plotted in subfigure b, are proportional to the surface coverage.

The temporal behaviour of the gas densities shows distinct trends for different species. $\text{N}_2\text{O}(\text{g})$ shows a typical, limited exponential increase during the plasma operation (phase I), followed by an exponential decrease once the plasma is switched off (phase II). The same can be stated for $\text{NO}(\text{g})$, although within a larger error margin. Such behaviour is expected when considering a constant source – i.e. the plasma – and the gradual diffusion-driven filling of the reactor volume.

In contrast, $\text{NO}_2(\text{g})$ and $\text{O}_3(\text{g})$ show sudden changes that indicate a varying production rate. $\text{O}_3(\text{g})$ spikes immediately after plasma ignition, then it decreases to a steady-state value of approximately $2.0e13 \text{ cm}^{-1}$. The initial overshoot is likely due to oxygen radicals sputtered from surfaces by the plasma, which promotes ozone formation. After continued plasma operation, this effect diminishes and $\text{O}_3(\text{g})$ likely equilibrates through diffusion. Similarly, the $\text{NO}_2(\text{g})$ density exhibits an initial overshoot as well, albeit to a smaller extent. Once the plasma is switched off, the $\text{NO}_2(\text{g})$ density spikes accompanied by a drop in the $\text{O}_3(\text{g})$ density. As such, the densities of $\text{O}_3(\text{g})$ and $\text{NO}_2(\text{g})$ are strongly connected.

Both nitrates are formed during plasma operation. They reach a steady-state value after 8 min. After switching off the plasma, $(\text{Fe-O})_2\text{NO}$ decreases and $\text{Fe-O}_2\text{NO}$ increase. This could indicate a relaxation to a weaker binding configuration at the surface. Finally, when the reactor is evacuated, the nitrates disappear alongside the removal of $\text{NO}_x(\text{g})$ species. This indicates the desorption of these nitrates. It seems that

the nitrates follow the gas species in time. This possible relation is further investigated in the next section, where $\text{NO}_x(\text{g})$ species are not directly formed by the plasma.

N_2 and O_2 Plasmas

The formation of NO_x surface groups during pure O_2 and N_2 plasma is studied next, where we alternated between a pure O_2 and N_2 plasma. Those were ignited for 20 min before evacuating the low-pressure compartment.

Figure 6a shows the absorption spectra near the end of plasma operation, where the pure O_2 plasma pretreatment step is plotted as well. The reference spectra were acquired prior to each plasma ignition. Therefore, the spectra reveal the change in chemical composition induced by the respective plasma step. The spectral positions of different surface groups are indicated using vertical grey dashed lines. The contributions of $\text{CO}_2(\text{g})$ are obscured by plotting this part more faintly. The rotational bands of ozone and nitrous oxide are fitted. The *residual* spectrum is the difference between the *measured* spectrum and these gas phase contributions. This leaves only surface contributions in the spectrum. The gas densities are plotted in Fig. 6b and c. There, the time stamp of the plotted spectra is highlighted using the black dashed lines.

Spectrum *I* agrees well with the O_2 plasma spectrum at 21 min of Fig. 3b, as they are both operated after the $\text{He}+\text{O}_2$ treatment. The spectrum shows the formation of iron(III) oxide and ozone and the removal of hydrocarbons, *M1*, and *M2*, i.e. carbon impurities.

Spectrum *II* shows the impact of operating a pure N_2 plasma. The following observations are made: the Fe_2O_3 band is not visible, instead iron oxyhydroxides FeOOH removal around 1130 cm^{-1} is observed while $\text{N}_2\text{O}(\text{g})$ is formed [29]. Likewise, the same can be said for the second nitrogen plasma of spectrum *IV*.

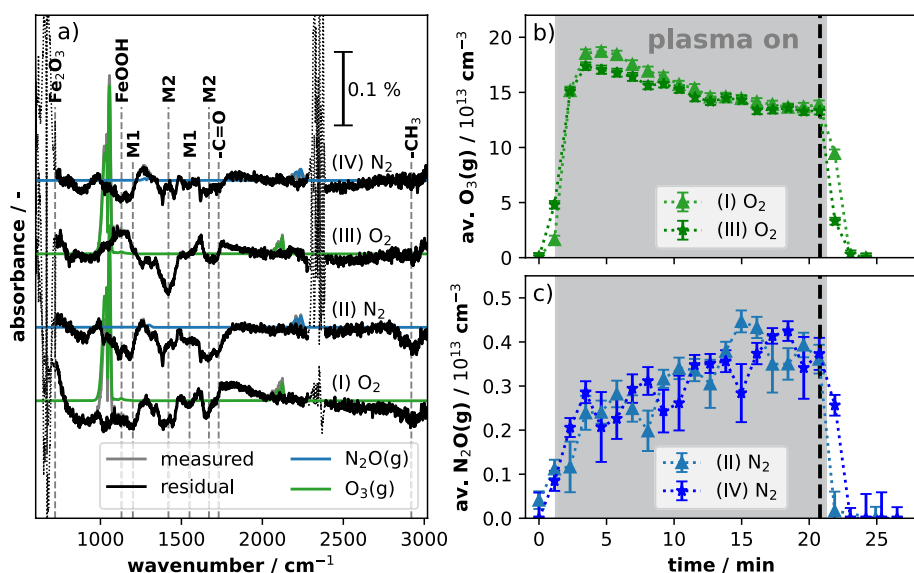


Fig. 6 IRRAS spectra obtained at 21 min for a pure O_2 (I), N_2 (II), O_2 (III), and N_2 (IV) plasmas (a). The $\text{O}_3(\text{g})$ (b) and $\text{N}_2\text{O}(\text{g})$ (c) densities are plotted for the pure O_2 and N_2 plasma, respectively

Spectrum III shows the impact of igniting a second O_2 plasma. It differs from the first oxygen plasma in the following aspects: the Fe_2O_3 signal is absent and $FeOOH$ formation around 1130 cm^{-1} is observed.

The iron(III)oxide band is not visible in spectra II, III, and IV. This could be obscured by the bending mode of $CO_2(g)$ around 620 cm^{-1} . Also, the typical iron oxide bands occur below 600 cm^{-1} [45]. This is below the spectral range of our MIR detector. Thus, minor changes in the oxidation in the toplayer of the substrate go undetected.

$FeOOH$ is formed in reactions with H atoms originating from the dissociation of impurities such as water or hydrocarbons, similar to O_2 plasma measurements in [29]. Such processes could be related to the ease of hydroxylation of iron oxides, as a partial pressure of 4×10^{-8} is already sufficient to trigger this process [46].

$N_2O(g)$ is only observed during the N_2 plasma cycles. It is likely formed by reactions of plasma-activated nitrogen with oxygen that is sputtered from the surface. This sputtering can explain the initial rapid rise in density. The Fe_2O_3 band is not affected because this band is not sensitive to small changes in the overall oxide layer.

The $O_3(g)$ density shows an initial overshoot similar to the $N_2:O_2$ plasma experiment shown in Fig. 5a, although the overshoot is smaller. This is attributed to the absence of $NO_x(g)$ species, which act as quenching partners for ozone. Similar to $N_2O(g)$, the initial overshoot is associated to sputtered oxygen species from the surface.

It is interesting to note that the IR signature for NO_x formation at the surface during the sequential exposure to the N_2 and O_2 plasma is almost absent. Following the process of chemical looping, one would expect that the exposure of an O-covered surface to a flux of N radicals during subsequent N_2 plasma treatment would lead to the formation of NO_x species at the surface. Yet, this is not observed. Therefore, it seems that nitrates are only formed in the presence of plasma-produced $NO_{1,2}(g)$ species.

Discussion

The formation of NO_x surface groups is studied in both a $N_2:O_2$ plasma and sequentially operating a pure O_2 and N_2 plasma. The $N_2:O_2$ plasma experiments show the formation of nitrates on the surface as well as the formation of $N_2O(g)$, $NO(g)$, $NO_2(g)$, and $O_3(g)$. The NO -to- NO_2 density ratio results from the balance between the oxidation of NO by O_3 and the reduction of NO_2 by O during the plasma operation [47, 48]. Once the plasma is turned off, the oxygen atoms quickly recombine to $O_2(g)$ and $O_3(g)$. Ozone exhibits a long lifetime and oxidises $NO(g)$ towards $NO_2(g)$. This gives a spike in the $NO_2(g)$ density when switching off the plasma.

Furthermore, the changes on the surface introduced by plasma operation are reversed after switching the plasma off. This is illustrated in Fig. 7. The pretreatment oxidises the top layer of the iron substrate. This gives the *plasma off* composition in this figure. Nitrates

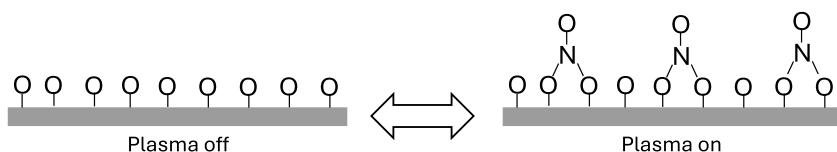


Fig. 7 A rough sketch of the change in the chemical composition of the iron substrate when igniting a $N_2:O_2$ plasma ignition

are formed once the plasma is operated, which is visualised in step *plasma on*. Afterwards, they desorb and the original *plasma off* surface composition is restored.

The nitrate formation is related to the plasma produced $\text{NO}(\text{g})$ and $\text{NO}_2(\text{g})$ species. This is evident from the differences between a $\text{N}_2:\text{O}_2$ plasma and sequentially operating a pure O_2 and N_2 plasma, where neither is observed. Also, the desorption of nitrates coincides with the removal of $\text{NO}_{1,2}(\text{g})$ species from the reactor volume.

Generally, nitrates are described as stable surface groups. High temperatures are required to desorb these groups, which typically starts at 200° [9, 21]. However, this is not observed in our low temperature experiments. Figure 5 shows that bridging bidentate nitrate decreases once the plasma is switched off. Similarly, evacuating the experiment compartment removes chelating bidentate nitrates as well. The temperature typically required to overcome the desorption barrier likely decreases as a result of plasma operation, e.g. due to plasma-produced $\text{O}_3(\text{g})$. Therefore, the immediate nitrate desorption in Fig. 5 is to be expected.

The absence of other nitrogen oxide surface groups is similar to the thermal equilibrium conditions for $\text{NO}_x(\text{g})$ adsorption in oxygen-rich environments. On metallic iron, the adsorption of $\text{NO}(\text{g})$ generally creates nitrosyls and NO^+ surface groups [9, 10, 33, 42, 43]. When $\text{NO}(\text{g})$ is adsorbed on a surface pre-dosed or co-adsorbed together with $\text{O}_2(\text{g})$, then the formation of nitrites and nitrates is observed at the cost of NO surface group formation. In a $\text{NO}_2(\text{g})$ atmosphere, nitrite/nitrate formation is the most dominant surface group. Yet, adsorbed N_2O_4 and NO are still observed [9]. The surface composition slightly changes when using a metal oxide substrate. Then, the formation of NO -groups is absent, and the spectra only show nitrites/nitrate formation [35, 49]. Hence, oxygen-rich environments favour the formation of nitrates. This is very similar to our conditions. Plasma-produced oxygen radicals are typically the dominant driver for chemical reactions in $\text{N}_2:\text{O}_2$ plasmas [47]. Therefore, the presence of oxygen in the plasma is the likely explanation of the absence of other nitrogen oxides on the surface, i.e. through quickly oxidising all adsorbed nitrogen oxides.

Finally, the nitrate production is associated with two plasma processes. First, the oxidising nature of O_2 containing plasmas ensures the full oxidation of any adsorbed nitrogen oxide. Second, only strongly bound surface groups survive when the substrate is in contact with the plasma. Weakly bound species are quickly removed by collisions with excited species from the plasma. Our experiment can be compared to DFT calculations that showed [50] that the adsorption of $\text{NO}(\text{g})$ on iron oxide and the subsequent desorption as $\text{NO}_2(\text{g})$ seems to be favoured. However, more experiments are needed to observe such trends.

Conclusion

The surface composition of an iron oxide foil in direct contact with an N_2 , O_2 , or $\text{N}_2:\text{O}_2$ plasma is studied using IRRAS in situ and in real-time. The reflection spectra show the formation of nitrates, iron(III)oxide, and the presence of various impurities. The oxidation degree does not rely on the N_2 to O_2 gas flow ratio, as even a pure nitrogen plasma is unable to reduce the surface.

Nitrates are formed by $\text{N}_2:\text{O}_2$ plasma exposure. It is argued that its formation is related to the adsorption of plasma-produced $\text{NO}_{1,2}(\text{g})$ species. The plasma creates reactive species such that other NO_x surface groups are readily oxidised towards nitrates. The results

presented in this article show that the formation of nitrates is possible when the surface is in direct contact with the plasma. This could potentially indicate the oxidation of NO(g) to NO₂(g) over the surface.

Future work will continue examining the formation of nitrates at different N₂/O₂ gas flow ratios. The NO to NO₂ ratio varies with changing this gas flow ratio, thereby elucidating the presumed differences in the adsorption of these NO_x species.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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